Hubbard architecture comparison: compilation, locality, and injection

Working research note: second refinement

22 September 2026

We refine the physical-qubit calculation and separate injected-state error from physical gate error in the factory model. The shared task remains the central doublon expectation after a half-filled Néel quench on an open square, with $U/J = 4$. The target is 1% relative error at the provisional signal $D_* = 0.1$, hence $\epsilon = 10^{-3}$, with 95% statistical confidence. We retain $\tau = 1$ and the shared step count $r = 4$ for the main comparison. The additional $r = 1$ calculations expose circuit locality; neither step count is certified against the allocated Trotter bias. No Hubbard dynamics curve is simulated.

1 Native gates and fermionic motion

We extract number-conserving two-mode hopping, fermionic Hadamard, and density blocks from the first compiled workload. Their noncommuting order is unchanged. Our physical gate set contains arbitrary $R_{ZZ}(\phi)$ and one-qubit pulses, each taking 10 ns. A two-qubit depolarizing fault has total probability $p = 10^{-3}$; a one-qubit fault has probability $p/10$. The idle channel has error p per 100 ns. These are explicit native-gate assumptions, not a characterization of a particular device.

Fermionic swaps provide a way to bring interacting modes together [1]. In the adjacent occupation basis, write

$$V(\theta) = e^{-i\theta(XX+YY)}, \qquad F_{\text{swap}} = (S \otimes S)V(\pi/4). \tag{1}$$

Thus $F_{\text{swap}}V(\theta)$ requires the same two variable-angle entanglers as $V(\theta + \pi/4)$, together with local phases. More generally, a number-conserving Gaussian gate is specified by an odd-sector matrix $G \in U(2)$; its vacuum amplitude is one and its double-occupancy amplitude is $\det G$. We decompose

$$G = e^{i\delta}R_z(\alpha)R_x(\beta)R_z(\gamma). \tag{2}$$

Opposite one-qubit Z rotations implement odd-sector R_z. Two native entanglers implement the odd-sector R_x through

$$U_x(\beta) = e^{-i\beta XX/4}e^{-i\beta YY/4}. \tag{3}$$

Equal final Z rotations supply the determinant. Products of Gaussian gates on the same pair therefore still require at most two R_{ZZ} gates. Density rotations commute with these pair gates and add at most one further R_{ZZ}.

We keep a live fermionic mode order and use only swaps adjacent in that order. A small lookahead chooses which endpoint moves. Pair operations can fuse through operations on disjoint wires. We select among four lookahead lengths using the local-PEC cost under the declared noise model. This is a heuristic compiler, not an optimal swap network. A separate implementation uses a two-root parity tree on the physical square grid; it preserves the complete Jordan–Wigner string and uses grid edges outside the original snake path.

The experimental comparison needs care. The Phasecraft/Quantinuum implementation uses all-to-all connectivity to make ordinary SWAP a relabeling operation; this lets an isolated fermionic swap

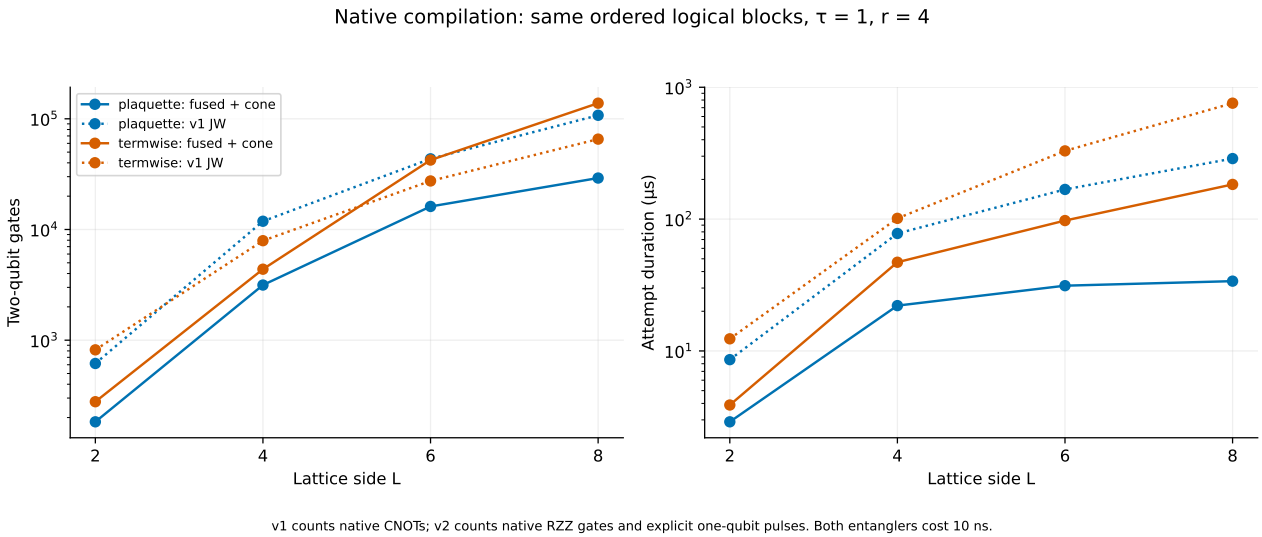


Figure 1: Actual CUDA-Q P0 counts and project-side ASAP durations at $\tau = 1$, $r = 4$. The v1 reference counts native CNOT gates; the refinement counts native R_{ZZ} gates and explicit one-qubit pulses. Disjoint gates may overlap. The primitive noise and scheduling assumptions, rather than a hardware backend, determine these durations.

use one entangler [2]. Our nearest-neighbor array must implement the mode motion. We therefore do not use its gate count as a direct calibration of the array's routing cost.

At $L = 4$, the plaquette circuit restricted to the sufficient observable cone uses 3149 native entanglers, compared with 11872 CNOTs in the original parity-ladder implementation. Its attempt duration is 22.07 μs, including 100 ns reset and 100 ns final readout. The native full-circuit version uses 3546 entanglers. Two-dimensional parity trees alone do not win: retaining long strings remains costly. At larger sizes, the greedy fused termwise route can also be worse in gate count than the original JW route. No compiler dominance is assumed.

2 What locality removes

All logical two-mode blocks are even in fermion operators. Blocks on disjoint fermionic supports therefore commute. Starting with the two observed modes, we traverse blocks backwards, retaining each intersecting block and adding its modes. Dropping the remaining blocks leaves the observable exactly unchanged for the selected circuit. For our occupation-basis initial state and diagonal occupation measurement, we can also compress out the untouched fermionic modes. This is an algebraic fermionic restriction, not a generic partial trace of interleaved JW qubits.

The resulting sufficient plaquette cone saturates at 36, 100, and 324 Hubbard sites for $r = 1, 2, 4$, respectively, in the interior cases checked. These counts follow $(4r + 2)^2$ for this stage ordering. The final plaquette layer first expands the support to a 2×2 tile. Each shifted tiling adds one site on each boundary; the reverse traversal of one complete step therefore reaches a 6×6 square. Each further step adds four sites to the side length, until a lattice boundary clips the support. In particular, the $r = 1$ circuits at $L = 8, 12, 16, 20$ each compile to exactly 2001 native entanglers on 72 modes. At $r = 4$, the cone still contains every mode of the $L = 4$ circuit, although some late blocks can be dropped.

The conservative dependency cone of the ordered termwise sweep reaches the whole array over the sizes checked, even at $r = 1$. A sequential product formula can propagate strict circuit dependencies through a long chain in one step. This does not establish a large necessary support for the observable

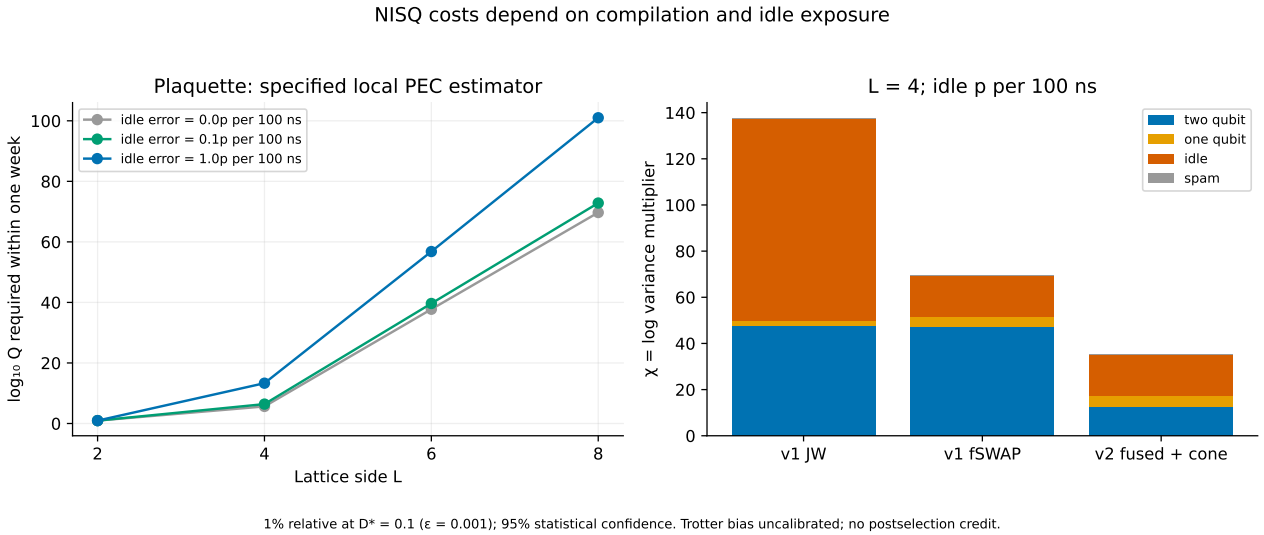


Figure 2: Conditional PEC costs and their exposure decomposition. The precision is $\epsilon = 0.001$, with half allocated to statistical error. The idle sensitivities keep the compiled circuit fixed. No postselection improvement is credited, and the reserved Trotter error has not been verified.

and does not contradict a finite-accuracy Hamiltonian light cone. Edge-colored schedules and finite-tolerance truncation are distinct further refinements. The same exact cone restriction is available to STAR and FT; it is not a NISQ advantage.

3 PEC exposure and idle time

For physical PEC, we retain only noise locations in the backward dependency cone of the compiled local measurement. CPTP channels outside this cone cannot change its expectation. Within it, the product inverse-channel construction is the same sufficient estimator as in the first note. If $s_k = 1 - 4^k p_k/(4^k - 1)$, the variance exposure of a k-qubit depolarizing location is

$$\log \gamma_k^2 = 2\log(1 + 2p_k/s_k), \qquad p_{\mathrm{idle}}(\Delta t) = \frac{3}{4}\left[1 - (1 - 4p/3)^{\Delta t/(100\,\mathrm{ns})}\right]. \tag{4}$$

The $L = 4$ plaquette exposure is

$$\chi = \log \Gamma^2 = 12.60_{\mathrm{2q}} + 4.89_{\mathrm{1q}} + 17.67_{\mathrm{idle}} + 0.14_{\mathrm{SPAM}} = 35.30. \tag{5}$$

At the nominal idle rate, the conditional week-cutoff area is approximately 2.1×10^{13} qubits for this estimator and circuit. Setting idle noise to zero, while retaining all gate errors and the same circuit, reduces it to 4.39×10^5 qubits. At one tenth of the nominal idle rate it is 2.56×10^6. These sensitivities show why the old conclusion based on 12000 CNOTs should not be promoted into a device-independent NISQ limit.

Preparing each product-state wire just before first use and reading it out after last use removes avoidable boundary idles, assuming no measurement crosstalk. We account for this separately. It removes only 16 idle qubit-layers from the local cone of the selected $L = 4$ plaquette implementation; nearly all its relevant idle exposure is internal to the circuit.

4 Postselection: exact acceptance, unresolved observable bias

We first implement selection on the two spin parities $P_\sigma = (-1)^{N_\sigma}$, not exact particle-number selection. For the compiled circuits, each of the three characters $P_\uparrow, P_\downarrow, P_\uparrow P_\downarrow$ remains a single Pauli under backwards propagation through every primitive. This is checked explicitly; the shortcut rejects a non-Clifford gate that would spread a character.

Each independent depolarizing location multiplies a character by s_k if its restriction to that location is nonidentity, and by one otherwise. Preparation and readout flips contribute $1 - 2p$ on the relevant wires. Let Λ_σ be the product of these factors, normalized to the ideal parity eigenvalue. The exact acceptance is then

$$A = \frac{1}{4}\left(1 + \Lambda_\uparrow + \Lambda_\downarrow + \Lambda_{\uparrow\downarrow}\right). \tag{6}$$

This derivation includes the primitive gate basis and the scheduled idle intervals. It assumes neither that errors occur at Gaussian-block boundaries nor that the noisy state is a classical occupation distribution.

At $L = 4$, we obtain $A = 0.2505$ for the refined plaquette circuit. The fully randomized parity limit is $1/4$, so a nonvanishing acceptance says little by itself about the retained state. If P_0 is the probability of no faults, the only state-independent conditional bias guarantee we currently use is

$$|D_{\text{selected}} - D_{\text{ideal}}| \le 1 - P_0/A. \tag{7}$$

It follows by separating the accepted no-fault component from all accepted faulty components, with $0 \le D \le 1$. At $L = 4$ this bound is 0.9997; even at $L = 2$ it is about 0.123. These loose sufficient bounds do not measure the actual bias, but they do not certify $\epsilon = 0.001$.

Exact spin-resolved particle-number selection is stronger and remains to be implemented for the noisy estimator. Its maximally mixed benchmark illustrates the same distinction: for $N = L^2$ sites at half filling,

$$A_{N_\uparrow = N_\downarrow = N/2} = \left[2^{-N}\binom{N}{N/2}\right]^2 \sim \frac{2}{\pi N}, \qquad D_{\text{selected}} = \frac{1}{4}. \tag{8}$$

Polynomial acceptance alone therefore does not restore an arbitrary target doublon expectation. We have not multiplied PEC overhead by $1/A$ and called the result a corrected estimator: combining PEC with conditioning requires its own numerator/denominator and bias analysis.

5 Injected-state error changes the hybrid result

The distillation input is an injected-state error p_{in}, separate from the physical circuit error p. For the retained leading-order serial 15-to-1 reference [3],

$$p_1 = 35p_{\text{in}}^3, \quad a_1 = 1 - 15p_{\text{in}}, \qquad p_2 = 35p_1^3, \quad a_2 = 1 - 15p_1. \tag{9}$$

We keep the same 12-tile one-level and 37-tile serial two-level factory references, replacing only the input error and corresponding acceptance. The physical surface-code error law still uses $p = 10^{-3}$.

Akahoshi et al. quote conventional injection coefficients 46/15 and 34/15, and use $46p/15$ in their FT comparison [4]. These coefficients depend on the injection circuit and its noise model; we have not recalibrated them for our asymmetric $p_{1q} = p/10$ convention. They define source-backed input-quality scenarios here. Using $p_{\text{in}} = 46p/15$ gives $p_1 \simeq 1.009 \times 10^{-6}$, about 28.8 times the old $35p^3$. At $L = 4$, 576 fixed-angle states then exceed the hybrid magic-error budget, even before its PEC amplification. Two-level distillation is required in the current model.

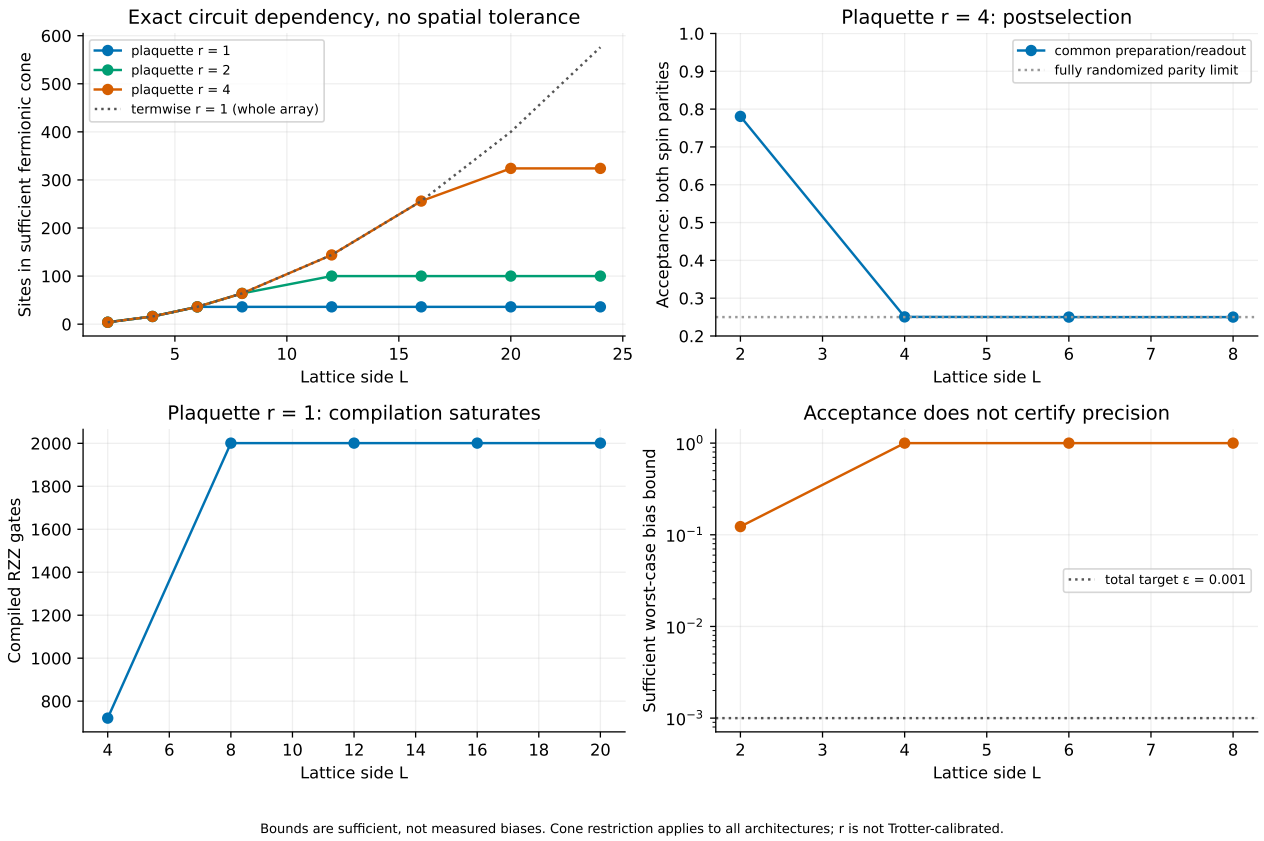


Figure 3: Sufficient circuit cones, compiled cost saturation at fixed r, and spin-parity postselection. The final panel is a worst-case bias upper bound, not a measured bias or a lower bound. The strict termwise dependency cone should not be confused with a finite-tolerance Hamiltonian light cone.

$L = 4$, $\epsilon = 0.001$	$p_{\text{in}} = p$	$p_{\text{in}} = 46p/15$
All-angle STAR area	166698	166698
Hybrid area	66150	669714
Hybrid distillation levels	1	2
FT area	13791250	13791250

Thus the earlier 2.52-fold hybrid improvement reverses at this point under conventional injection and the serial factory reference. The benefit survives at $L = 8$ at about 7.3-fold. Replica rounding is not the whole explanation: relaxing replica integrality for the selected configurations gives about 1.30×10^5 qubits for all-angle STAR and 5.25×10^5 for the corrected hybrid at $L = 4$.

Precision and injection quality must be varied jointly. At $\epsilon = 0.01$, one-level FT passes with $p_{\text{in}} = p$, but still fails with $p_{\text{in}} = 46p/15$. The hybrid passes with one level at this looser precision in both cases. This sensitivity holds the more accurate compiled FT words fixed, so re-synthesis could further reduce the looser-precision cost. The actual v1 fit was 70 T gates per dynamic rotation; counting the fixed gates as well gives 72 per dynamic rotation. GridSynth gave 66 dynamic T gates.

There is a further architecture choice: Akahoshi's postselected prepared-state error is $2p/15 + O(p^2)$, while the $4p/15$ rotation error includes repeat-until-success consumption [4]. Feeding the better

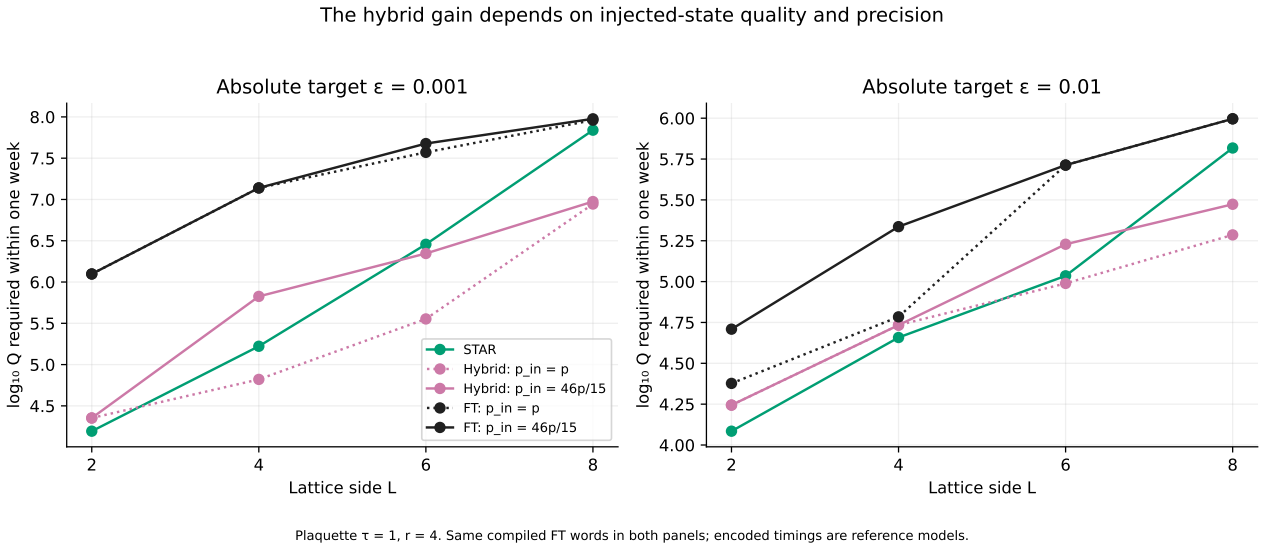


Figure 4: Factory switching changes the hybrid result. Both panels use the compiled FT words for $\epsilon = 0.001$; the right panel isolates looser error allocation and sampling. Timing, tile counts, logical-error fits, and serial factory supply remain the declared encoded reference models.

prepared state into distillation can change both FT and hybrid factory levels. We include $2p/15$ as a conditional sensitivity, but have not charged a compiled implementation of its injection acceptance and supply. A fair optimized comparison must allow FT to use improved injection too; a single input-error coefficient is not an architecture law.

6 Scope of the completed refinement

The new native circuits compile in CUDA-Q Logical without using the rejected encoded backend schedule. Local matrix identities verify Gaussian fusion, fermionic permutations, parity trees, and cone compression. An independent two-mode density-channel calculation checks parity acceptance with every native fault location, including idle and boundary timing. These are local circuit and channel checks, not a many-body dynamics calibration.

The remaining NISQ work is a better topology-aware ordering, exact particle-number selection and its residual observable bias, and a calibrated combined mitigation estimator. STAR injection/growth/retry schedules and the encoded scheduling reproducer remain the next architecture tasks. The C9 commuting-Pauli sensitivity is retained; its use as a STAR default still requires a residual-channel and branch-dependence argument for those gadgets. Trotter calibration and qubitization retain their later position.

References

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